Supplemental Results Discussion for MLPerf Storage v2.0

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MLCommons Supplemental Discussion

MLPerf Storage v2.0 Results Discussion

The submitting organizations provided the following 300 word descriptions as a supplement to help the public understand their MLCommons® MLPerf® storage v2.0 submissions and results. The statements **do not reflect the opinions or views of MLCommons.**

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Supplemental Key Principles - all supplementals must adhere to these key principles. All submitters should publicly review supplementals on Wednesday, 7/23, and feedback may be given for corrections if these principles are not followed.

- Be positive! Focus on why your submission is great.
- No negativity or comparisons, even implied comparisons. Don't knock competitors.
- Help audiences understand the positive impact on the broader community.

Please note that all materials from MLCommons must be neutral and provide a fair environment

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Alluxio

Alluxio Enterprise AI 3.6 Achieves Breakthrough MLPerf Storage 2025 Results

Alluxio, the leading data orchestration platform for AI and analytics, proudly participates in the MLPerf® Storage v2.0 benchmark with Alluxio Enterprise AI 3.6. Powered by Distributed Cache and a unified namespace, it accelerates AI workloads with memory-speed data access across diverse storage systems, ensuring high GPU utilization with minimal complexity.

Originating at UC Berkeley's AMPLab, Alluxio serves nine of the top ten internet companies, offering a robust data orchestration layer for AI and analytics. Its memory-first architecture enables seamless data access in hybrid and multi-cloud environments, forming a cornerstone for scalable AI infrastructure.

The MLPerf Storage benchmark evaluates single-node and multi-node performance, with the multi-node category reflecting large-scale AI training environments. It measures how effectively storage saturates GPUs across multiple nodes during standardized machine learning tasks.

Highlights from the 2025 Submission:

1. **Support for up to 128 H100 GPUs** on ResNet-50 with a massive dataset, linearly scaling seamlessly across 100 nodes.
2. **Over-99.57% GPU utilization**, maximizing resource efficiency across all workloads.
3. **Robust scalability on CosmoFlow**, supporting up to 20 H100 GPUs with a massive dataset across multiple nodes.

These results demonstrate Alluxio Enterprise AI 3.6's exceptional performance in demanding multi-node AI workloads. Leveraging NVMe SSDs and distributed caching, Alluxio ensures GPUs remain fully utilized with fast, reliable data access, while asynchronous checkpointing enhances training efficiency and resilience.

Alluxio accelerates model development by providing a unified POSIX interface to disparate data sources, reducing data engineering overhead and enabling seamless access across clouds and on-premises environments. This empowers organizations to scale AI workloads cost-effectively, supporting computer vision, natural language processing, and analytics.

By participating in MLPerf Storage, Alluxio reinforces its commitment to transparent, community-driven benchmarking, delivering reliable insights for AI infrastructure innovation.

About Alluxio

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Founded in 2015 at UC Berkeley's AMPLab, Alluxio bridges compute frameworks like PyTorch and TensorFlow with diverse storage systems, delivering memory-speed data access. With deployments at hundreds of organizations, Alluxio drives efficient AI and analytics in hybrid and multi-cloud environments.

Follow Alluxio: LinkedIn, X, and YouTube.

LinkedIn, X, and YouTube.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Argonne National Laboratory

Argonne's DAOS and Lustre Excel in MLPerf Storage Benchmark for Large-Scale AI

Training massive AI models, such as the 405B and 1T Llama3, presents a significant I/O challenge during the critical process of checkpointing. This process of periodically saving the model's state is essential for fault tolerance, protecting potentially weeks of computational progress from hardware or software failures. At the Argonne Leadership Computing Facility (ALCF), we leveraged the MLPerf Storage v2.0 benchmark on the Aurora exascale supercomputer to validate storage solutions capable of meeting this intense demand and minimizing idle compute time.

Two primary parallel file systems on Aurora were evaluated:

- **DAOS:** The benchmark utilized a 128-server node subpartition of Aurora's 230 PB DAOS system, an open-source object store designed for extreme performance on NVM and NVMe SSDs to support AI and HPC workloads
- **Lustre:** The 100 PB 'Flare' file system, a robust solution built on an HPE ClusterStor E1000 platform with an aggregate transfer rate of 650 GB/s.

Benchmark Highlights: Our MLPerf testing yielded outstanding I/O throughput, essential for efficient, large-scale AI training:

- **DAOS** delivered exceptional performance, reaching nearly **1 TB/s write throughput** and 600 GB/s read throughput for the Llama3 405B model. This real-world speed allows a full model checkpoint to be saved in **under 10 seconds**.
- **Lustre provided strong, consistent performance,** with sustained write and read speeds between **400-600 GB/second**, confirming its capability to operate effectively at near-peak capacity for demanding workloads.

Paving the Way for Exascale AI: These benchmark results validate the architectural strength of Aurora's storage ecosystem. They confirm that both DAOS and Lustre provide the scalable, reliable, and high-performance solutions necessary to support the most complex and data-intensive AI applications. By overcoming critical storage bottlenecks, Argonne is enabling researchers to push the boundaries of science and accelerate discovery on its exascale systems.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

DDN

DDN AI400X3 Delivers Breakthrough MLPerf Storage 2025 Results

DDN, the AI Data Intelligence Company, is proud to participate in the MLPerf® Storage v2.0 benchmark with its latest high-performance storage solution: the DDN AI400X3. This next-generation appliance is powered by the DDN EXAScaler parallel file system, accelerating the world's most demanding AI workloads at scale with high performance density and minimal power consumption.

DDN has delivered exceptional performance across intensive AI and HPC workloads for decades. Since 2016, NVIDIA has relied exclusively on DDN to power its internal clusters, underscoring DDN's leadership in scalable, high-performance AI storage systems.

DDN's results in the MLPerf Storage benchmark's multi-node training workloads are especially relevant to AI practitioners, as it better reflects how storage systems support AI training workloads at scale. The benchmark measures how many GPUs can be saturated by a single storage appliance during standardized machine learning tasks across multiple client nodes.

Highlights from the 2025 submission:

- **120+ GB/s sustained read throughput for 3D U-Net H100 training**
- **Support for up to 640 simulated H100 GPUs on ResNet-50**
- **2× performance improvement on CosmoFlow with the new AI400X3**

These results demonstrate the AI400X3's ability to deliver consistently high performance across diverse AI workloads, even under heavy, multi-node training demands.

The AI400X3 accelerates model training by keeping GPUs fully utilized with fast, reliable data access and supports frequent checkpointing without disrupting performance, reducing costs while improving training efficiency and resilience.

DDN helps organizations reduce time-to-insight and scale confidently across domains like medical imaging, scientific computing, and computer vision. Designed for performance, efficiency, reliability, and simplicity, the AI400X3 supports sustainable AI growth.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

By participating in MLPerf Storage, DDN reinforces its commitment to transparent, community-driven benchmarking and provides organizations with trusted data to guide infrastructure decisions and AI strategies.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

ExponTech

ExponTech WADP Delivers Incredible Performance & Efficiency in MLPerf Storage v2.0 Running Entirely on DPUs

ExponTech WADP is a next-gen software-defined storage platform providing block, file, object, and KV protocols on a unified engine. Demonstrating exceptional results in MLPerf Storage v2.0, the WADP software ran entirely within 4 NVIDIA BlueField-3 DPUs installed in a single 2U AIC JBOF housing 26 ScaleFlux CSD5000 NVMe SSDs. Notably, WADP is highly flexible and also runs efficiently on standard commodity servers.

WADP delivers excellent performance for both mission-critical and AI workloads, making it a true unified platform for enterprise AI across all scenarios.

Key Highlights:

Ultra-Dense & Efficient: Single 2U JBOF with DPUs as remote storage nodes.

Highly Scalable: WADP software manages to hundreds of JBOF appliances with its distributed software architecture

Incredible Performance in MLPerf Storage v2.0 testing:

ResNet-50 (Single Client): 240 A100 GPUs @ 23.6 GiB/s; 128 H100 GPUs @ 24 GiB/s.

ResNet-50 (Dual Clients): 400 A100 GPUs @ 38 GiB/s; 200 H100 GPUs @ 37 GiB/s.

Llama 3 8B: checkpoint write@16.9 GiB/s, checkpoint read@11.4 GiB/s.

Extreme I/O Efficiency: Achieved stable 96% AU during many rounds testing

Validated Scalable Network: Four NVIDIA Spectrum-X switches formed a 2-layer RoCE network, confirming excellent congestion handling. With the 2-layer network and scalable distributed WADP storage software, the solution can be scaled out to sustain massive large scale AI training and inference

Enhanced Efficiency: ScaleFlux CSD5000 SSDs provide transparent data compression/decompression, multiplying raw capacity and efficiency without performance impact or extra resource consumption.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

FarmGPU

FarmGPU Unleashes 121 GB/s Beast: Single Intel® Xeon® Server showcases single-node storage performance for AI workloads

FarmGPU is a GPU cloud providing scalable compute to enterprises and startups, with specialized storage offerings tailored for AI workloads.

FarmGPU is proud to announce results in the MLPerf® Storage v2.0 benchmark, submitting a **single-node configuration** that delivered strong throughput and efficiency. The server is using Intel® Xeon® 6 with P-core and a class-leading 136 lanes of PCIe Gen 5 bandwidth. FarmGPU optimized the entire local storage stack with optimal RAID, filesystem, kernel, and memory tuning. The barrier to entry for AI data storage systems can be over a dozen storage and compute nodes; we wanted to showcase that even a single storage server can handle many GPUs if tuned properly.

The single-socket Supermicro SYS-212H-TN storage server, powered by an Intel® Xeon® 6781P CPU was sponsored by Intel. The server was architected with additional PCIe lanes for storage use and equipped with 24x Solidigm D7-PS1010 SSDs. This system achieved exceptional throughput as a single compute node hosting storage:

3D U-Net: 121 GB/s with 40 simulated H100s at > 95% AU

CosmoFlow: 19 GB/s with 36 simulated H100s at > 70% AU

ResNet-50: 15.64 GB/s with 84 simulated H100s at > 91% AU

Additionally, our second submission, featuring a Supermicro SSG-121E-NES24R with two Intel® Xeon® Platinum 8450H CPUs and 24x FADU Echo Gen5 E1.S SSDs, demonstrated excellent performance on a previous gen system with the same single-node configuration.

These results underscore FarmGPU's expertise and innovative approach to AI infrastructure, proving that superior performance can be achieved efficiently at the granular level before requiring expensive, large-scale deployments.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Hammerspace

Hammerspace Demonstrates the High Performance and Unmatched Efficiency of Tier 0 for GPU Computing in the MLPerf® Storage v2.0 Benchmark

Hammerspace is the Data Platform for AI Anywhere, enabling organizations to activate the AI-ready infrastructure they already own. We're excited to announce our MLPerf Storage v2.0 results, highlighting the performance, scalability, and efficiency possible with our cutting-edge approach.

Testing focused on Hammerspace's Tier 0 implementation, which integrates the NVMe storage already present in compute servers into the Hammerspace shared file system. This parallel global file system unifies NVMe across multiple GPU server nodes, external file and object storage, and cloud instances. It uses pNFSv4.2 FlexFiles and locality-aware orchestration to intelligently place and protect data. The result is globally shared data accessible at local PCIe bus speeds—maximizing AI pipeline performance.

Hammerspace submitted results in the Closed division for the 3D U-Net workload simulating H100 accelerators. In that context:

- Hammerspace supported 140 simulated H100 accelerators, with bandwidth of over 80GiB/s per client for single and multiple client configurations.
- This result was achieved with exceptional hardware efficiency, requiring just one additional 1U node for the metadata server. All other operations ran locally within the benchmark clients. No dedicated storage servers were needed, as data storage devices were housed directly within the clients.
- Running the test using one, three, and five clients (the largest tested) showed linear scalability in the number of supported accelerators as client count increased.
- No proprietary client software or kernel patches were needed on the clients. Hammerspace is standards-based, leveraging the high-performance capabilities of pNFS built into standard Linux.
- Standard Ethernet was used, proving that achieving high performance for AI doesn't require deploying a special, additional network type.

Hammerspace Tier 0 lets organizations jump-start AI projects using the infrastructure they already own, saving rack U, power, cooling, and capital.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

HPE

HPE Cray Supercomputing Storage Systems E2000 was benchmarked for the first time and highlights HPE's high performance computing (HPC) results, specifically for critical scientific workloads in medical imaging and cosmology parameter prediction training tasks. HPE Cray Supercomputing Storage Systems E2000 is the successor to HPE Cray ClusterStor E1000 Storage Systems, which has seen extraordinary adoption in the supercomputing industry. Specifically, it is used in:

- All three of the world's verified exascale supercomputers with a combined usable storage capacity of nearly 1.3 exabytes across these three systems
- Six of the world's fastest top 10 supercomputers, per the TOP500 list

HPE Cray Supercomputing Storage Systems E2000 features exceptional price/performance and reliability, and more than doubles the I/O performance as compared with the previous generation with the use of new storage controllers, making it the ideal storage solution for supercomputers.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

IBM

IBM Storage Scale Delivers Real-World Performance

Training large language models is a complex and resource-intensive process; in an ideal world, you would have a cluster dedicated and tuned to just one task. IBM takes a more real-world, cost-effective approach, engineering AI clusters to simultaneously support diverse workloads, including multiple training workloads at different scales, as well as fine-tuning, evaluation, and even inference.

That's why we're thrilled to share these MLPerf Storage benchmark results, which include reading and writing checkpointing workloads utilizing the Llama Herd (8B, 70B, 405B, and 1T) family of models.

Highlights include:

- Llama 3 1T: Scale delivered read bandwidth of 656.7 GiB/s and write bandwidth of 412.6 GiB/s or approx. 23 seconds to load and 37 seconds to save, respectively.

- Llama3 405B: Scale delivered read bandwidth of 624.7 GiB/s and write bandwidth of 384.7 GiB/s or approx. 8.5 seconds to load and 14 seconds to save, respectively.

What's especially impressive is that we allotted just 4% of our Blue Vela H100 cluster (https://arxiv.org/abs/2407.05467) and ran the checkpointing benchmark workloads on up to 32 nodes, while a mixture of IBM Research workloads fully utilized the remaining 96%.

As the parameter count of LLMs continues to grow, so do their checkpoint sizes. A failure of a checkpoint write erases all the work done by the computation since the last checkpoint, which can constitute a significant expense. While 32 compute nodes are fewer than the number realistically necessary for pre-training larger models efficiently, this test demonstrates IBM Storage Scale's ability to deliver on the most critical portion of a training workload – scalability.

These results highlight our commitment to pushing the boundaries of storage performance for evolving AI workloads across industries in real-world scenarios. We look forward to continuing our leadership in new AI use cases and benchmarks, providing the highest performance and efficiency to our customers.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Jinan Institute of Supercomputing Technology

Jinan Institute of Supercomputing Technology (JNIST) and Huawei Technologies Co., Ltd have achieved outstanding collaborative results in MLPERF® Storage v2.0 Benchmark Suite

As a leading high-performance computing research institute with large-scale AI computing power in China, JNIST has adopted Huawei all-flash high-performance AI storage (OceanStor A800 and OceanStor A600) to provide multiple ML workloads, comprehensively validating the performance of Huawei AI storage in complex AI scenarios.

Data shows that in the 3D U-Net workload, a single OceanStor A800 appliance provided 698 GiB/s of stable bandwidth to support simulation training on 255 H100 GPUs. OceanStor A600 provided 108 GiB/s bandwidth per U to meet the training demands of 38 H100 GPUs, demonstrating the outstanding performance density of Huawei OceanStor A series storage. In checkpointing test cases, Huawei AI storage achieved 40.2 GiB/s read bandwidth and 20.5 GiB/s write bandwidth for Llama3-8B single-client with eight simulated GPUs. Moreover, in the Llama 3 70B test case with eight clients and 64 simulated GPUs, Huawei AI storage achieved 195.3 GiB/s read and 114.7 GiB/s write bandwidth. These results were enabled by Huawei's innovative data-control plane separation architecture and cluster scaling technology, enabling exceptional data supply capabilities under hybrid workloads in AI scenarios.

These real-world tests verify that the OceanStor A series storage delivers high bandwidth and low latency for AI training scenarios. Its performance density and scalability provided a reliable data foundation for large-scale distributed training. The joint solution by JNIST and Huawei achieved efficient utilization of GPU/NPU computing resources and provides a reference for optimizing storage architectures in AI computing centers, research platforms, and beyond, helping improve large AI model training efficiency. Moving forward, both parties will further drive innovation in storage technology and promote the integrated development of high-performance computing and AI applications.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

JuiceData

JuiceFS is a high-performance cloud-native distributed file system, ideally suited for multi-cloud environments. As a user-space distributed file system, it delivered impressive results in MLPerf® Storage v2.0 Benchmark.

We conducted several tests across various workloads on a single host, and the results are as follows:

- 3D U-Net: A 3D image segmentation model performing consecutive dataset reads achieved 92% accelerator utilization on 40 H100 GPUs with 10 host.
- ResNet-50: This well-known CNN model, primarily used in computer vision and performing random reads from datasets, achieved 94.9% accelerator utilization on 500 H100 GPUs with 10 host.
- CosmoFlow: Analyzing N-body cosmology simulation data to predict physical parameters of the universe, this 3D convolutional neural network reached 74% accelerator utilization on 100 H100 GPUs with 10 host.

JuiceFS' architecture separates "data" and "metadata" storage. Data is stored in object storage, while metadata is stored in our in-house developed, high-performance metadata engine. Distributed caching has significantly enhanced I/O performance, enabling JuiceFS to satisfy the requirements of high-performance scenarios.

JuiceFS uses Ethernet as the communication medium for its distributed caching, offering a bandwidth of approximately 12 GB/s. Although Ethernet's performance is lower compared to networks like InfiniBand, potentially limiting JuiceFS' performance, its ease of maintenance and cost-efficiency ensure that Ethernet remains the mainstream network choice for most enterprises.

Compared to kernel-mode file systems, JuiceFS may exhibit slightly lower performance due to the increased overhead from system calls and data copying. However, it significantly enhances system security through its user-space operations.

Notably, JuiceFS is one of the few solutions that utilize object storage for data storage. Furthermore, the Enterprise Edition of JuiceFS is competitively priced, providing a cost-effective AI storage solution to users. This is particularly advantageous in large-scale AI training scenarios where traditional all-flash and kernel-mode parallel file systems are associated with high costs.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Overall, as a user-space distributed file system, JuiceFS offers AI enterprises a cost-effective and high-performance storage solution. Its cloud-native feature, combined with POSIX compatibility, integrates seamlessly into the modern AI ecosystem.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Kingston Technology

Kingston Technology's DC3000ME NVMe SSD series delivered impressive results on the MLPerf® Storage v2.0 benchmark suite, demonstrating peak performance in AI training and checkpointing tasks, allowing storage architects to maximize GPU utilization of their AI clusters without IO bottlenecks. We tested all capacities of the DC3000ME on two modern PCIe Gen5 platforms and one PCIe Gen4 system to highlight the performance benefits of using PCIe Gen5 to maximize AI training throughput, while also providing a clear and understandable picture of the performance expectation of using DC3000ME in PCIe Gen4 systems.

Training

Kingston's DC3000ME test results confirmed robust accelerator capabilities for all the training workloads:

· **ResNet-50**: DC3000ME maintained 127 A100 accelerators at 91.6% accelerator utilization on the SR-665V3 system with 11,601 MiB/s sustained throughput. H100 setups provided 109,527 samples/sec with 11,977 MiB/s bandwidth with 65 accelerators at 93.4% AU, significantly higher than the 90% threshold requirement.

· **3D U-Net**: Peak bandwidth achieved 12,591 MiB/s for 9 A100 accelerators at 91.2% AU on the SYS-621H-TN12R system. H100 performance delivered 11,933 MiB/s with 4 accelerators sustaining 98.8% accelerator utilization, showing great efficiency for medical imaging workloads.

· **CosmoFlow**: The drives sustained 19 H100 accelerators at 73.4% AU with 10,383 MiB/s throughput for cosmological simulation workloads. A100 configurations maintained 23 accelerators at 7,878 MiB/s while surpassing the 70% AU threshold.

These results highlight the significant impact of PCIe Gen5's enhanced bandwidth for checkpointing efficiency. The SEDC3000ME/15T3 drive delivered 11.1 GiB/s load bandwidth and 10.3 GiB/s save performance for Llama 3 models. Llama 3 405B operations finished within 8.6 seconds for loads and 9.6 seconds for saves, a significant improvement over Gen4 checkpointing (up to 4x improvement).

Kingston's DC3000ME series (3.84TB, 7.68TB, 15.36TB) uses enterprise-grade, 3D NAND flash technology and an enterprise-level controller, with intelligent firmware effectively tuned for peak performance, sustained Quality of Service, and reliable latency.

MLPerf Storage v2.0 certification ensures these drives satisfy the rigorous accelerator utilization criteria necessary for production AI workloads, offering storage architects qualified performance metrics for large-scale GPU cluster rollouts.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

KIOXIA

In the constantly evolving world of machine learning (ML) and artificial intelligence (AI), the requirements for keeping modern GPUs constantly utilized is challenging. Traditional training requires high capacity in order to hold large datasets that can include text, high resolution images, and videos. In addition, regular checkpointing is also required in order to ensure that model training is not interrupted in case of any failures. Storage subsystems must be able to maintain high read and write throughput to service many GPU threads concurrently, where each thread is serviced quickly and maintain low latency from the storage backend.

KIOXIA addresses customer needs through our technological leadership in the field of flash memory and SSDs, and through our broad portfolio of products. KIOXIA SSD products are built using our BiCS FLASH™ 3D flash memory, delivering optimized SSD products and solutions for the most demanding ML/AI workloads.

KIOXIA CM7-R Series is a read intensive SSD that is optimized to support a broad range of applications including machine learning and artificial intelligence. Built on PCIe® 5.0 and NVMe™ 2.0 technology, the CM7 Series E3.S SSDs offer high capacity points up to 15.36 TB deliver excellent performance of up to 14,000 MB/s (sequential read) and 6,750 MB/s (sequential write).

A single CM7-R Series 7.68 TB E3.S SSD was utilized and was able to show the following results:

· Serviced four (4) H100 GPUs concurrently showcasing read bandwidths on average of 11.3 GB/s for the 3D U-Net benchmark

· Serviced sixty (60) H100 GPUs concurrently showcasing read bandwidths on average of 10.7 GB/s for the ResNet-50 benchmark.

· Serviced fourteen (14) H100 GPUs concurrently showcasing read bandwidths on average of 9.7 GB/s for the CosmoFlow benchmark.

· Loading checkpoints with an average throughput of 6.6 GB/s and saving checkpoints with an average throughput of 7.6 GB/s for the Llama 3 8B model.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Lightbits Labs

Lightbits Labs Showcases the Advantages of NVMe over TCP Block Storage in MLPerf Storage v2.0 Benchmarks

Lightbits Labs delivered standout performance in the MLPerf® Storage v2.0 benchmarks, reinforcing its position as a leader in high-performance, software-defined block storage for AI/ML workloads. The company's disaggregated architecture consistently met the intensive I/O demands of modern AI using just **three commodity storage servers**, highlighting the performance, scalability, and efficiency of its NVMe® over TCP solution.

Benchmark Results:

- **3D U-Net**
    - A100: 41.24 GiB/sec total throughput; 469.2 MiB/sec per accelerator per storage node (6 clients)
    - H100: 41.26 GiB/sec; 939 MiB/sec per accelerator per storage node (5 clients)
- **ResNet-50**
    - A100: 39.32 GiB/sec; 31.07 MiB/sec per accelerator per storage node (6 clients)
    - H100: 41.71 GiB/sec; 59.32 MiB/sec per accelerator per storage node
- **CosmoFlow**
    - A100: 16.64 GiB/sec; 126.24 MiB/sec per accelerator per storage node
    - H100: 21.84 GiB/sec; 191.16 MiB/sec per accelerator per storage node (3 clients)

These results reflect Lightbits' ability to deliver high throughput with minimal infrastructure, making it a compelling option for organizations seeking performance and efficiency at scale.

As the inventor of **NVMe over TCP**, Lightbits integrates the protocol natively into its software, enabling ultra-low latency and exceptional throughput over standard Ethernet. This composable approach allows customers to scale storage independently from compute, streamlining infrastructure for dynamic AI workloads.

A founding participant in the MLCommons storage benchmark, Lightbits promotes transparency by publicly sharing performance data with customers and competitors alike.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

To learn more, visit www.lightbitslabs.com and follow Lightbits on LinkedIn, X, Facebook, Instagram, and YouTube.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

MangoBoost

MangoBoost is a DPU and system software company aiming to offload network and storage tasks to improve the performance, scalability and efficiency of modern datacenters. We are excited to introduce **Mango StorageBoost. StorageBoost is a storage solution utilizing DPUs to offload NVMe/TCP to the hardware, allowing the DPU to function as an initiator device on the host side and a target device on the storage system side.**

In this MLPerf Storage 2.0 submission, we evaluated our DPU solution with 2x200G NTT-NTI (2-card) systems on the Intel Xeon 6438N (64-core) and an AMD EPYC 9965 (384-core) hosts, each utilizing Mango StorageBoost NVMe/TCP Initiator DPU connecting to a single fabric-attached block storage equipped with Mango StorageBoost NVMe/TCP Target DPU. With this setup, we showcase the scalability and cost-efficiency of our DPU in handling MLPerf storage traffic per unit of fabric-attached storage.

For the training benchmark, our DPU solutions demonstrate that our DPU achieved line-rate performance at full 400Gbps bandwidth on both A100 and H100 3D U-Net benchmark, achieving a maximum of 31 A100 GPUs at 1.42 GiB/s throughput per accelerator (94.7% AU), and 15 H100 GPUs at 2.87 GiB/s throughput per accelerator (92.9% AU). These results show how our DPU offloading allows the storage system to fully utilize the 400Gbps network and achieve high performance efficiency.

These results highlight how our DPUs enable AI storage systems—including traditional and JBoF systems—to scale efficiently. **MangoBoost's suite of DPU and hardware solutions provides complementary hardware acceleration and AI enterprise software solutions for training, finetuning, and inference serving, significantly reducing system overhead in modern AI/ML workloads.** Aside from the Mango StorageBoost Initiator and Mango StorageBoost Target showcased in this round, Mango GPUBoost offloads the RoCEv2 RDMA stack at high connection scalability. Mango NetworkBoost offloads web-serving and TCP stacks, dramatically reducing CPU utilization. Mango LLMBoost AI Enterprise software provides turnkey solutions for AI deployment at scale.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Micron

Micron is proud to announce results for the Micron 7.68TB 7600 and 15.36TB 9550 NVMe™ SSDs. Results show that these drives excel in AI workloads where extreme speed and scalability are paramount.

For the Training workloads, a single 7.68TB Micron 7600 SSD can optimally service up to:

- **65x H100 accelerators training ResNet-50 at 11.7GB/s**
- **22x H100 accelerators training CosmoFlow at 11.8GB/s**
- **4x H100 accelerators training 3D U-Net at 11.2GB/s**

And a single 15.36TB Micron 9550 SSD can optimally service up to:

- **72x H100 accelerators training ResNet-50 at 13.1GB/s**
- **26x H100 accelerators training CosmoFlow at 14.2GB/s**
- **4x H100 accelerators training 3D U-Net at 11.2GB/s**

For Checkpointing workloads, a Micron 7600 and 9550 SSD support saving and loading checkpoints for Llama 3 of all tested model sizes: 8B, 70B, 405B, and 1T.

A single 7.68TB Micron 7600 SSD can optimally:

- **Save checkpoints at up to 10.5GB/s**
- **Load checkpoints at up to 9.9GB/s**

And a single 15.36TB Micron 9550 SSD can optimally:

- **Save checkpoints at up to 13.5GB/s**
- **Load checkpoints at up to 13.1GB/s**

These outstanding results underscore the fact that the Micron 7600 and 9550 SSDs offer the performance required to support many accelerators. Unlike many SSDs tailored for specific read or write-intensive use cases, the 7600 and 9550 SSDs deliver high performance with balanced read and write capabilities that are ideal for AI workloads.

With capacities up to 30.72TB and supporting a wide array of form factors, the Micron 7600 and 9550 SSDs also deliver exceptional energy efficiency, security, and storage density for AI workloads. For more information, visit Micron's website: www.micron.com/7600 and www.micron.com/9550.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

NEW H3C Technologies Co., Ltd

H3C Group, a leading provider of digital and AI solutions, is committed to becoming a trusted partner in its customers' business innovation and digital transformation journeys. As a core enterprise of the New Unisplendour Group, H3C leverages its comprehensive strategy across the entire industry chain of "cloud-network-security-computing-storage-endpoint," possessing full-spectrum capabilities in digital infrastructure, including computing, storage, networking, 5G, security, and terminals.

H3C Storage, with the strategic vision of "Intelligent by Design, Empowering Intelligence," is dedicated to building intelligent data infrastructure for the future. In 2024, H3C introduced the Polaris X20000 series, a high-performance distributed converged storage solution designed for AI computing scenarios. Powered by a newly developed, proprietary storage engine and leveraging erasure coding technology, Polaris X20000 achieves higher storage efficiency compared to traditional replication methods. The system incorporates a RoCE-based network to deliver ultra-low latency and high bandwidth, effectively eliminating data transfer bottlenecks and enabling single-node bandwidth up to 150 GB/s. Designed to meet the demands of AI and HPC workloads, it offers exceptional performance for massive data processing.

Furthermore, Polaris X20000 integrates block, file, object, and HDFS protocols into a unified storage platform, enabling seamless data flow across the entire AI pipeline—from data collection and model training to distribution—without the need for data migration. This provides AI applications with stable, efficient, and cost-effective storage support.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Nutanix

Progress You Can Measure: Nutanix Unified Storage Pushes AI Performance Farther

Nutanix continues to raise the bar in AI infrastructure, delivering exceptional results in the 2025 MLPerf® Storage benchmark. This year, the Nutanix Unified Storage (NUS) solution once again achieved 2x performance gains for AI applications over 2024—using half the resources compared to last year. A single Nutanix cluster demonstrated the ability to service up to 2,312 GPU accelerators, achieving a 130% per-node performance improvement. These results underscore the platform's advanced capabilities.

Nutanix Unified Storage again excelled in key MLPerf categories like image classification and image segmentation. Performance was powered by advanced features including NFS over RDMA and SR-IOV support—technologies designed to reduce latency and bypass CPU bottlenecks - helping NUS to deliver on its ability to sustain heavy AI workloads. With no proprietary client-side drivers, Nutanix, with its Unified Storage platform, delivered these results via our software-defined NAS file server using standard ethernet networking.

Beyond performance, Nutanix delivers enterprise-grade resiliency, security, and simplicity. It supports deployment across on-premises, edge, and cloud environments, enabling organizations to scale AI initiatives with confidence—from centralized training to edge inference.

These results reinforce Nutanix's commitment to delivering a future-ready storage platform that empowers IT teams to accelerate AI outcomes—without complexity getting in the way. We're proud to partner with MLCommons and support their mission to advance benchmarking best practices in the rapidly evolving AI landscape.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Oracle

OCI Lustre Filesystem: Enabling Efficient, Scalable AI Workflows

Oracle Cloud Infrastructure (OCI) Lustre Filesystem (OCI LFS) delivers a reliable, performant, and scalable storage solution for AI and HPC workloads in the cloud. Built on the open-source Lustre platform, OCI LFS empowers customers to address modern machine learning and data needs with a fully managed, high-throughput storage layer.

For our MLPerf Storage submission, we deployed OCI LFS with 540.8 TB (125 MB/s/TB performance tier), rated at 67,600 MB/s aggregate throughput. Clients mounted OCI LFS leveraging a standard **50 Gbps** Ethernet frontend network. (**Note**: newer OCI GPU compute shapes offer frontend networks ranging from 100 to 400 Gbps.)

Benchmark Results:

- Training performance peaked at over 60,000 MiB/s on ResNet-50 (H100 GPUs) and over 60,000 MiB/s on 3D U-Net (A100 and H100 GPUs)—all nearly saturating the system's rated throughput of 67,600 MB/s. CosmoFlow benchmarks achieved over 42,000 MiB/s for A100 GPUs and over 39,500 MiB/s for H100 GPUs. These results highlight OCI LFS's ability for high throughput across a range of AI training scenarios and configurations.
- OCI LFS achieved strong throughput results in checkpointing runs. Checkpointing throughput measured 4.8 GiB/s (5.1 GB/s) for writes and 4.9 GiB/s (5.2 GB/s) for reads from a single 50 Gbps client on Llama3-8b, and reached 58.3 GiB/s (62.5 GB/s) for writes and 62 GiB/s (66.5 GB/s) for reads on Llama3-405b—almost saturating the system's throughput capability. These results illustrate OCI LFS's support for demanding language model workloads at both small and large deployment sizes.

These benchmark results highlight that OCI Lustre Filesystem delivers dependable performance for real-world AI workloads on cloud infrastructure. With easy deployment, seamless scaling, cloud-native security, and tight OCI integration, OCI LFS is a compelling choice for your AI training and inference workloads.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Quanta Cloud Technology

As AI workloads continue to grow in complexity and data intensity, the need for high-throughput and low-latency storage solutions becomes increasingly important. In the MLPerf Storage 2.0 submission, QCT and Solidigm jointly demonstrate how carefully integrated system design and advanced SSD technologies can improve storage performance and data handling for AI pipelines.

QCT submitted results using its QuantaGrid D54X-1U server platform, a 1U system purpose-built for high-performance computing, data-centric workloads, and AI infrastructure. With a well-balanced CPU-to-storage ratio, versatile I/O options, and thermal-optimized architecture, the D54X-1U provides a robust foundation for storage-intensive workloads across AI pipeline.

Two configurations were tested using the same server platform:

- One featuring Solidigm D7-PS1010 NVMe SSDs, designed to deliver predictable low-latency performance, particularly effective for metadata-intensive AI operations.
- The other using Solidigm D5-P5336 NVMe SSDs, offering high capacity and outstanding streaming read throughput—key for efficient access to AI model and dataset files.

This collaboration highlights how thoughtful system design and next-generation storage technology can drive measurable improvements in AI performance. The results from QCT and Solidigm showcase a production-ready, scalable solution that enables data-driven enterprises to unlock faster insights, streamline AI pipelines, and improve storage efficiency.

The MLPerf Storage benchmark provides a valuable, community-driven framework to evaluate end-to-end storage performance under realistic machine learning conditions. QCT and Solidigm are proud to support the MLCommons community and remain committed to advancing innovations that enhance the efficiency and effectiveness of AI development and deployment.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Samsung Electronics Co., Ltd. (Memory Business)

Samsung's NVMe™ SSD excels at traditional performance standards while also delivering high throughput and capacity optimized for such demanding environments. As AI architectures continue to evolve rapidly, the ability to proactively address these emerging performance demands will become increasingly important - and Samsung remains at the forefront of researching these demands and implementing solutions to stay ahead of the needs of AI accelerator technology.

Samsung submitted two sets of results: Training results based on H100 GPU and Checkpoint results based on various parameters on the CLOSED division of the MLPerf® Storage v2.0 Benchmark using PM1753 NVMe™ SSD.

For PM1753, all results for training workloads of CosmoFlow, ResNet-50, and 3D U-Net show a read performance of about 11.4 GB/s. It also shows a write performance of up to 11.8 GB/s in checkpointing workloads.

These results demonstrate that Samsung’s NVMe™ SSDs provide stable, high-speed performance in AI applications based on large datasets.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Sandisk

Sandisk delivers innovative Flash solutions and advanced memory technologies that meet people and businesses at the intersection of their aspirations and the moment, enabling them to keep moving and pushing possibilities forward.

In the era of data-driven enterprises, achieving high-performance and low-latency data storage solutions is essential to drive business success. The rapid emergence of artificial intelligence (AI) as a primary workload in modern data centers is driving a paradigm shift in power and performance requirements. Accelerated compute resources, such as GPUs, are exponentially increasing power demands, pushing data center architects to deliver higher performance.

The Sandisk SN861 SSD is Sandisk's next-generation NVMe™ SSD, designed specifically for enterprise data centers and high-performance workloads. The Sandisk SN861 is built with advanced NAND technology, delivering:

- High throughput: Ensures faster data access and processing.
- Low latency: Provides quick response times for time-sensitive applications.
- Endurance and reliability: Optimized for continuous, 24/7 enterprise workloads.

Sandisk submitted results for 3D U-Net, with H100 GPUs within the CLOSED division for MLPerf Storage 2.0 benchmarks:

- For SN861, results show that for the 3D U-Net training workload we could support up to 4 GPU’s (with over 92% AU), and achieved a training throughput of 79.2 Samples/sec; and an I/O throughput of 11078.0300 MB/s.
- The test was specifically conducted on SN861 U.2 15.36TB drive.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Simplyblock

Simplyblock Showcases Excellent Throughput per Accelerator in MLPerf Storage v2.0

Simplyblock, a cloud-native data platform, delivered strong performance in the MLPerf® Storage v2.0 benchmark, demonstrating its ability to meet the demanding I/O needs of modern AI training workloads with top-of-class efficiency.

The storage benchmark results showcase simplyblock's ability to sustain **high throughput per accelerator** while maintaining high GPU utilization, **reaching up to 65 Gbit/s of read bandwidth per client**, helping ensure that compute resources are fully utilized across a range of representative AI workloads.

The benchmark ran on a minimal **three-node storage cluster** using standard **Google Cloud Compute Engine VMs** (`c4a-standard-64-lssd`), leveraging a **Kubernetes-native architecture** with **NVMe-over-TCP**, and requiring **no host-side modifications** to bring high-performance storage capabilities to mainstream cloud users.

Simplyblock executed all three available workloads in CLOSED configuration, without benchmark tuning.

- In **ResNet-50**, simplyblock achieved **180 MiB/s per accelerator** across configurations using **44 and 76 H100 GPUs**, accessed by just one or two clients, respectively. The system sustained over **125,000 samples/s**, fully utilizing GPU capacity at scale.
- In **CosmoFlow**, it achieved **530 MiB/s per accelerator** across **30 H100 GPUs**, demonstrating readiness for bandwidth-intensive scientific workloads.
- In **3D U-Net**, the system delivered **3 GiB/s per accelerator** on just **four H100 GPUs**, meeting both bandwidth and latency requirements for unstructured data.

"Scalable AI training demands storage that performs as well as compute, and simplyblock proves it's up to the task," said Rob Pankow, CEO of Simplyblock. *"This benchmark validates our belief that today, with the right tooling, everyone can have access to high-performance AI infrastructure on demand, and easily surpass legacy systems' bottlenecks while reducing cost and complexity."*

The Simplyblock Platform, natively built on NVMe/TCP, enables enterprises to take full control of their data sovereignty and application performance, while empowering AI teams to scale with confidence, ensuring every GPU cycle is maximized.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

TTA (Telecommunications Technology Association)

The Telecommunications Technology Association (TTA) is a non-profit organization established in 1988, offering testing, certification, and standardization services for Artificial Intelligence (AI). TTA plays a key role in ensuring the safety, reliability, and interoperability of AI technologies by validating AI performance, evaluating compliance, and developing technical standards that support trustworthy AI deployment across industries.

TTA also supports global marketing and market share expansion for Korean server/storage HW/SW vendors. TTA recognizes the value of publishing performance results through platforms like MLCommons, and on this occasion, we test the storage server ExaStor manufactured by Gluesys Co, Ltd.

Gluesys, founded in 2000, is a leading Korean enterprise specializing in storage solutions and data management software. With a strong track record in financial, public, healthcare, and telecommunications sectors, the company provides high-performance storage systems to over 1,000 clients.

Its flagship product, ExaStor, is a high-performance scale-out storage system based on the Lustre parallel file system, optimized for AI and HPC workloads. It delivers fast, parallel access to data across storage nodes, supports GPU-direct memory access to reduce CPU load, and minimizes latency through Lustre/NFS over RDMA. The system can be applied across diverse fields such as meteorology, media, life sciences, and AI research.

TTA conducted performance evaluations using two Gluesys storage systems based on Lustre, equipped with A100 and H100 simulation accelerators. The tests measured training performance for models such as 3D U-Net and ResNet-50, as well as checkpointing performance for the Llama 3 70B model. Through this work, TTA aims to contribute meaningful data points to the MLPerf community and help others understand the practical storage considerations for future AI workloads. We believe that open benchmarking and active participation in collaborative initiatives such as MLPerf are key to fostering innovation and ensuring broad community benefit.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

UBIX TECHNOLOGY CO., LTD.

UBIX Delivered Outstanding Performance in the MLPerf® Storage v2.0 Benchmarks

UBIX Technology Co., Ltd. is a high-tech enterprise focused on innovative storage products and solutions. Through innovations in system architecture, and software implementation, the company has developed **UBIXFS**, a high-performance distributed file system with full intellectual property rights, and launched products such as **distributed all-flash storage, distributed massive storage, and AI Computing Appliance**.

UbiPower 18000 distributed All-flash storage, relying on an innovative two-layer storage pool architecture, with RDMA high-speed transmission and storage virtualization, builds a hierarchical and highly efficient resource pool. This architecture accelerates the overall performance of the system to a higher level, and delivers excellent benchmark performance. **UBIX's innovative AI storage products** have been commercially deployed in multiple supercomputing and AI data centers, demonstrating excellent performance in scenarios such as data preprocessing, massive data access, and large-scale checkpoint read/write operations. These products provide high-performance storage services for scientific research, industrial HPC, and model development and training in text, video, and multi-modal large models.

During the testing, **UBIX** successfully completed all test items based on strategic partner **Yinfeng** Intelligent Computing Technology Co., Ltd.'s premium AI computing infrastructure.

In this MLPerf test, a total of 10 benchmark scenarios were defined, including 6 training and 4 checkpoint tasks. **UBIX** deployed 3 storage nodes and participated **in all 10 scenarios**. And **achieved outstanding performance in the following 7 scenarios, including**:

Training:

- **ResNet-50**: served **3120** of A100 accelerators, at **280.77GiB/s**.
- **ResNet-50**: served **2160** of H100 accelerators, at **374.57GiB/s**.
- **CosmoFlow**: served **608** of A100 accelerators, at **226.44GiB/s**.
- **CosmoFlow**: served **528** of H100 accelerators, at **273.21GiB/s**.
- **3D U-Net**: served **336** of A100 accelerators, at **455.05GiB/s**.

Checkpoint:

- **Llama 3 1T**: Delivered **54.7 GiB/s** of read bandwidth and **36.3 GiB/s** of write bandwidth per Storage System RU.
- **Llama 3 405B**: Delivered **50.5 GiB/s** of read bandwidth and **36GiB/s** of write bandwidth per Storage System RU.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

Western Digital

Western Digital's MLPerf Results Demonstrate Exceptional AI Storage Scalability and Throughput in partnership with PEAK:AIO and Kioxia

Western Digital is proud to announce its MLPerf® Storage v2.0 submission, highlighting the real-world performance of OpenFlex™ Data24 4000 Series NVMe-oF™ Storage Platform utilizing KIOXIA CM7-V NVMe SSD's, and PEAK:AIO AI Data Server. The results showcase how our architecture enables accelerated GPU performance with exceptional density and efficiency for modern AI workloads.

Our system was submitted in the CLOSED division, focusing on realistic deployment scenarios involving multiple client nodes accessing a single storage target. The benchmark measures how effectively the system serves AI workloads that stress different aspects of storage I/O, including throughput and concurrency, across various deep learning models.

- 3D U-Net Workload
    - The Western Digital OpenFlex Data24 achieved sustained read throughput of 99.2 GiB/s, saturating 36 simulated GPUs across 3 physical compute nodes showing the EBOF's ability to handle bandwidth-intensive, high-parallelism training tasks with ease.
    - With PEAK:AIO AI Data Server the OpenFlex Data24 was able to deliver 59.6 GiB/s, saturating 22 simulated GPUs from a single head server and single client node.

ResNet-50 Workload

- The Western Digital OpenFlex Data24 delivers optimal performance across 186 simulated GPUs and 3 client nodes, with an outstanding GPU-to-drive ratio that reflects the platform's efficient use of physical media.
- With PEAK:AIO AI Data Server the OpenFlex Data24 was able to saturate 52 simulated GPUs from a single head server and single client node.

These results demonstrate how Western Digital's disaggregated architecture can help maximize GPU utilization while minimizing footprint and complexity. The OpenFlex Data24 delivers

near-saturation performance across demanding AI benchmarks, both standalone and with a single PEAK:AIO AI Data Server appliance, translating to faster time-to-results and reduced infrastructure sprawl.

Supplemental Results Discussion for MLPerf Storage v2.0

This information is under embargo until 8/4/25 8:00AM PT

YanRong

YanRong Delivers Breakthrough Performance in MLPerf Storage v2.0 Benchmark

YanRong Tech, a leading provider of AI storage, has long been dedicated to building high-performance, reliable infrastructure for AI and data-intensive workloads. Building on our strong performance in last year's MLPerf® Storage v1.0, we have once again proven our capability in accelerating large-scale AI workloads and maximizing GPU utilization—achieving outstanding results in the latest MLPerf® Storage v2.0 benchmark.

At the heart of our achievement is YRCloudFile, a high-performance distributed parallel file system specifically designed to meet the demanding I/O needs of AI model training, inference, and HPC environments. Powered by YRCloudFile, our F9000X all-flash storage appliance delivers exceptional bandwidth, ultra-low latency, and linear scalability. In MLPerf® Storage v2.0, a 3-node F9000X cluster—the minimum viable scale for distributed storage—demonstrated superb performance in the UNet3D test, achieving 513 GB/s of bandwidth on commodity hardware.

Highlights of our MLPerf v2.0 results include:

- **3D U-Net**: 87 GB/s with 30 simulated H100 accelerators (single client); scaling to 513 GB/s with 176 accelerators (8 clients).
- **CosmoFlow**: 38 GB/s with 68 simulated H100 accelerators (single client); scaling to 137 GB/s with 243 accelerators (9 clients).
- **ResNet-50**: 42 GB/s with 220 simulated H100 accelerators (single client); scaling to 259 GB/s with 1404 accelerators (9 clients).

These results demonstrate YRCloudFile's ability to scale efficiently under increasing GPU and client workloads. With fully distributed metadata and data architecture and YanRong's Multi-Channel technology, the system maximizes bandwidth across InfiniBand and RoCE networks—delivering the high throughput and low latency needed for large-scale AI training, checkpoints, LLM inference, and GPU cloud workloads.

As AI workloads grow in complexity and scale, YanRong remains committed to advancing performance, scalability, and efficiency in AI infrastructure. We are proud to contribute to the MLCommons® community and look forward to continued collaboration in driving storage innovation for next-generation AI applications.